

Yimin (Michael) Wang

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EDUCATION

Wake Forest University

August 2022 – May 2026

B.S. in Computer Science and B.S. in Applied Statistics

Winston Salem, North Carolina

GPA: 3.87 | ACM Member | Upsilon Pi Epsilon member | Dean's List 2022-2026 | John W. Sawyer Prize winner

Relevant Coursework: Introduction to Regression and Data Science, Probability, Applied Generalized Linear Models, Introduction to High Performance Computing, Data Structures and Algorithms, Computer System I, Computer System II, Neural Networks & Deep Learning, Numerical Methods, Reinforcement Learning, Large Language Model, Computer Security

RESEARCH EXPERIENCE

Interpretable and Robust AI Research Group, Wake Forest University

May 2025 – Present

Undergraduate Research Assistant advised and funded by Dr. Fan Yang

Winston Salem, NC

Efficient Shapley Estimation across Tabular, Image, and Graph Data in LLM

- Designed a new framework for efficient Shapley value estimation with mathematical proofs of *unbiasedness* and *faithfulness*; leverages data priors to reduce variance and cost.
- Implemented a general, black-box compatible pipeline with data-type priors (antithetic grouping for tabular, random cropping for images, single-hop sampling for graphs).
- Tabular (Data Shapley for LLM training): extended dataset-level valuation to GPT-2 (small), identifying and filtering high-quality instances to accelerate convergence and improve downstream performance.
- Image/Video: performed pixel-level Shapley analysis across frames to generate time-sensitivity reports (e.g: detecting moments of human appearance)
- Graph/Mechanistic Interpretability: discovered functional subgraphs within an LLM that most contribute to a target output—without gradients or model retraining—advancing efficiency and interpretability.
- Extended a Shapley-value-based attribution method to token-level explanations for long-context LLM prompts, quantifying how instruction vs. content tokens each influence the model's generated output and highlighting the most decision-driving input segments

DataMine Research Group, Wake Forest University

May 2023 – August 2025

Undergraduate Research Assistant advised by Dr. Natalia Khuri

Winston Salem, NC

Data Centric Artificial Intelligence for Environmental Toxicology

- Evaluated different data-centric approaches, including coresets, on model's performance by performing paired t-test
- Implemented and tested a Genetic Algorithm(GA) for data set coreset selection problem
- Perfomed grid search for hyperparameter for GA by using parallel processing
- Collaborated with WFU School of Medicine to develop novel approaches for the analysis of biomedical datasets

Cluster analysis and Data integration tools for scRNA-seq data

- Contributed to the "scrnabench" Github Package
- Implemented and optimized a Multi-objective Genetic Algorithm for clustering high-dimensional and noisy datasets
- Implemented R scripts to retrieve, annotate, and clean scRNA-seq data to train a eXtreme Gradient Boosting model to predict the cell type of scRNA-seq data

Multi-Objective Coreset Selection for Data-Centric QSAR Modeling

- Visualize results for different implemented coreset selection methods with their minority ratio and selection time by using seaborn and matplotlib
- Conduct the experiment for different chemical data using different coreset selection methods and evaluate the performance for classification models

AWARDS AND GRANTS

- Wake Forest Research Fellowship, The Wake Forest URECA Center, 2024 summer research **May 10 2024 – July 3 2024**
- HPC Immersion Grant, SC23 HPC immersion committee, traveling, registration, and expenses **November 11 – 17, 2023**
- 2nd Place, ASA DataFest 2025, Wake Forest University — Recognized for outstanding data analysis and creative insight in a 48-hour national data science competition **April 4– 6, 2025**
- Computing Research Association (CRA) Undergraduate Research Training (URT) — Completed the training program and received an official certificate of completion. **September - December, 2025**
- IndySCC (Student Cluster Competition) — Completed the full competition program, including cluster setup/configuration, benchmarking, and performance optimization of HPC workloads **November 14 - 16, 2025**
- Honorable Mention, CRA Outstanding Undergraduate Researcher Award 2026, Computing Research Association **December, 2025**
- John W. Sawyer Prize in Computer Science, Wake Forest University; awarded to the most outstanding graduating senior in Computer Science and accompanied by a \$500 scholarship **May, 2026**

PUBLICATIONS

- K. Zhao, **Y. Wang**, O. Hernandez Segura, and N. Khuri, “Multi-Objective Evolutionary Optimization for the Discovery of Structurally Diverse and Active Compounds” [**Accepted, ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB 2026)**]
- **Y. Wang** et al., “DUET: A Dual-Game Shapley Mechanism for Understanding Large Language Models” [**Manuscript in preparation for submission to ICML 2026**]

ORAL AND POSTER PRESENTATIONS

- “Data-Centric Artificial Intelligence for Environmental Toxicology”, oral presentation, GLBIO 2025, Great Lakes Bioinformatics Conference, Minneapolis, MN, May 12, 2025

COMPETENCY

Research: Evolutionary multi-objective optimization, Algorithmic instability and fairness, Genetic Algorithms for cluster analysis, Environmental Toxicology, Data-Centric AI

Programming: Python, C, R, Java, R Markdown

Toolkits: Pytorch, Tensorflow, Scikit-learn, rdkit, PyGAD, xgboost, seaborn, matplotlib

PROFESSIONAL SERVICE AND EXTRACURRICULAR

Undergraduate Research Assistant (Paid Position) **August 2025 – Present**

Undergraduate Research Tutor, Department of Computer Science, Wake Forest University **September 2025 – Present**

WFU Chinese Men’s Basketball Team **October 2022 – Present**

WFU Chinese Students and Scholars Association - Sports Dept **September 2022 – Present**